

Problem Set 2

Due on Tuesday August 10 at 11:59 PM (if you want feedback) [It is fine to hand in solutions to selected problems.]

1. Assume that $f : [0, 3] \rightarrow \mathbb{R}$ is integrable and satisfies

$$f(x+1) = \frac{1}{2}f(x) \quad \text{for all } x \in [0, 2].$$

Given that $\int_0^1 f(x)dx = 1$, determine $\int_0^3 f(x)dx$.

2. Assume that f is differentiable on $[0, 1]$ and satisfies $f(0) = \frac{1}{2}$, $f(1) = 1$, and

$$f'(x) = (f(x))^2 \quad \text{for all } x \in [0, 1].$$

Evaluate $\int_0^1 f(x)^4 dx$.

3. For each improper integral, determine whether or not it converges. (You do not need to evaluate the integral.)

(a) $\int_0^\infty e^{-x^4} dx$

(b) $\int_1^\infty \frac{1+e^{-x}}{x} dx$

(c) $\int_1^\infty \frac{x^2}{\sqrt{1+x^6}} dx$

(d) $\int_0^\infty \frac{x}{\sqrt{1+x^6}} dx$

(e) $\int_{-\infty}^\infty e^{-\sqrt{|x|}} dx$

(f) $\int_{-\infty}^\infty e^{-x^3} dx$

4. Find all values of p for which the improper integral below converges and evaluate the integral for those values of p .

$$\int_e^\infty \frac{1}{x(\ln x)^p} dx$$

5. Find all values of $p \in \mathbb{R}$ for which the improper integral

$$\int_e^\infty \frac{(\ln x)^p}{x^2} dx$$

converges.

6. For each choice of the function F below, determine whether or not the improper integral

$$\int_0^\infty F(x) dx$$

converges.

(a) $F(x) = \int_x^\infty \frac{dt}{\sqrt{t^3 + 1}}, \quad x \geq 0$

(b) $F(x) = \int_0^x e^{-t^2} dt, \quad x \geq 0$

(c) $F(x) = \int_x^\infty e^{-t^2} dt \quad x \geq 0$